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Molecular Characterization of *Escherichia coli* Causing Community-Acquired Urinary Tract Infections in a Suburban Area in West Bengal, India: Dispersal of Multidrug-Resistant Epidemic ST131 Clone Carrying CTX-M-1

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Abstract

Antimicrobial resistance (AMR) imperils global public health overlooking the epidemiological situation of community-acquired infections, especially in low- and middle-income countries. We conducted this study in a suburban community in West Bengal, India, to comprehend the AMR and bacterial forensics of *Escherichia coli* causing community-acquired urinary tract infections (CAUTIs). Antibiograms, β-lactamases, plasmid-mediated quinolone resistance genes, and plasmids were investigated to explore AMR. Bacterial forensics was assessed using phylogrouping, genetic fingerprinting, and strain typing. Antibiograms revealed that > 75% of isolates had a multiple-antibiotic resistance (MAR) index of > 0.2. Fosfomycin remained 100% effective as an empiric drug, followed by nitrofurantoin (> 75%), aminoglycosides (> 65%), and co-trimoxazole (> 50%). Multidrug-resistant (MDR) was frequent (> 75%), with < 5% being extensively drug-resistant, including two colistin-resistant isolates. Piperacillin/tazobactam showed promising activity against the ESBL-producing isolates (> 70%) mostly carrying CTX-M-15 and TEM-1 (> 50%), which declined in the presence of OXA-1 (> 20%). TEM-253 and 254, two novel alleles, were assigned. Two of the thirteen NDM-5-producing carbapenem-resistant isolates co-harbored OXA-48. Among fluoroquinolone-resistant isolates (> 80%), *aac(6')-lb-cr* (> 20%) was the most common, followed by *qnrS* (> 10%). Plasmid of > 212 kb was prevalent (> 60%), especially in MDR isolates. IncF was the sole plasmid in > 85% of isolates, with > 90% having a MAR index > 0.2. Phylogroup B2 was prevalent (> 35%), primarily associated with the ESBL-producing ST131 clone (> 80%). Carbapenem-resistant isolates belonged to phylogroup A/sequence-type361, C/ST2851, D/ST405, and F/ST648. ESBL-producing isolates were largely clonally related. This maiden molecular study reveals the infiltration of MDR epidemic clones in a suburban community in eastern India, guiding genomic surveillance for strategic management of AMR at the national level.

Introduction

Urinary tract infections (UTIs) are the most common in outpatients and the second most prevalent in hospitalized patients, causing cystitis (lower UTI, bladder infection), pyelonephritis (upper UTI, kidney infection), and even life-threatening urosepsis [1]. It affects approximately 405 million people and kills nearly 0.23 million globally, resulting in 5.2 million disability-adjusted life years (DALYs) in 2019 [1, 2]. Uropathogenic *Escherichia coli* (UPEC), which belongs to extra-intestinal pathogenic *Escherichia coli* (ExPEC), causes 70–95% of UTI cases ascending from urethra to bladder to ureter to kidney to bloodstream, among which 80–90% account for uncomplicated community-acquired UTIs (CAUTIs), which often require hospitalization with severe complications commonly in women [1, 3]. In low- and middle-income countries (LMICs), including India, uncomplicated CAUTI infections are empirically managed in outpatient departments (OPDs) using oral antibiotics such as co-trimoxazole, nitrofurantoin, and fosfomycin

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to reduce progression in the absence of urine culture findings [4, 5]. Emerging multidrug-resistant (MDR) UPEC due to the fast dissemination of antibiotic resistance genes (ARGs) such as extended spectrum β-lactamases (ESBLs) and plasmid-mediated quinolone resistance (PMQR) restricts the global use of third-generation cephalosporins (3GCs) and fluoroquinolones approaching for last-line antibiotics like carbapenems, especially in South Asia [5–7]. In 2024, the World Health Organization (WHO) flagged 3GC-resistant *Enterobacterales* as a critical priority pathogen [8]. Antimicrobial resistance (AMR) is a global public health threat that has a significant impact on India, posing challenges to clinical management and threatening an epidemic situation due to over-the-counter antibiotic access, self-medication, overcrowding, cross-infections, poor healthcare infrastructure, awareness, regulatory issues, and commercial promotion by doctors [1]. Cross-border trafficking of international epidemic clones increases the risk of severe infections and community outbreaks [9, 10]. The blending of AMR and microbial forensics supports clinicians with antibiograms, microbiologists with resistance mechanisms, and epidemiologists with strain types, fostering prudent public health interventions [11]. Comprehensive molecular investigations on nosocomial isolates are available to understand the ever-changing dynamics of AMR in clinical settings, but the situation in communities in India is overlooked [4, 7, 12, 13]. *Escherichia coli* serve as the bioindicator of AMR due to their broad adaptability to diverse ecological niches, including humans, animals, and the environment [9, 14]. Investigation of UTIs through a common pathogen like *Escherichia coli* reveals the situation of AMR in communities. Hence, we captured local antibiograms, β-lactamases, PMQRs, plasmid profiles, phylogroups, genetic fingerprints, and strain types of *Escherichia coli,* causing CAUTI to inscribe the situation of AMR in a suburban community in West Bengal, India.

Materials and Methods

Study Design and Bacterial Isolates

This cross-sectional observational single-site study was conducted on CAUTI by *Escherichia coli* at the department of Biomedical Laboratory Science and Management, Vidyasagar University (BMLVU), in collaboration with the department of Clinical Microbiology, Spandan Multispeciality Hospital (SMH), Midnapore, from September 2021 to June 2023 (Fig. 1). Midnapore is a multiethnic populated suburban area and livestock and dairy farming hub in the Jungal Mahal districts of southern West Bengal, about

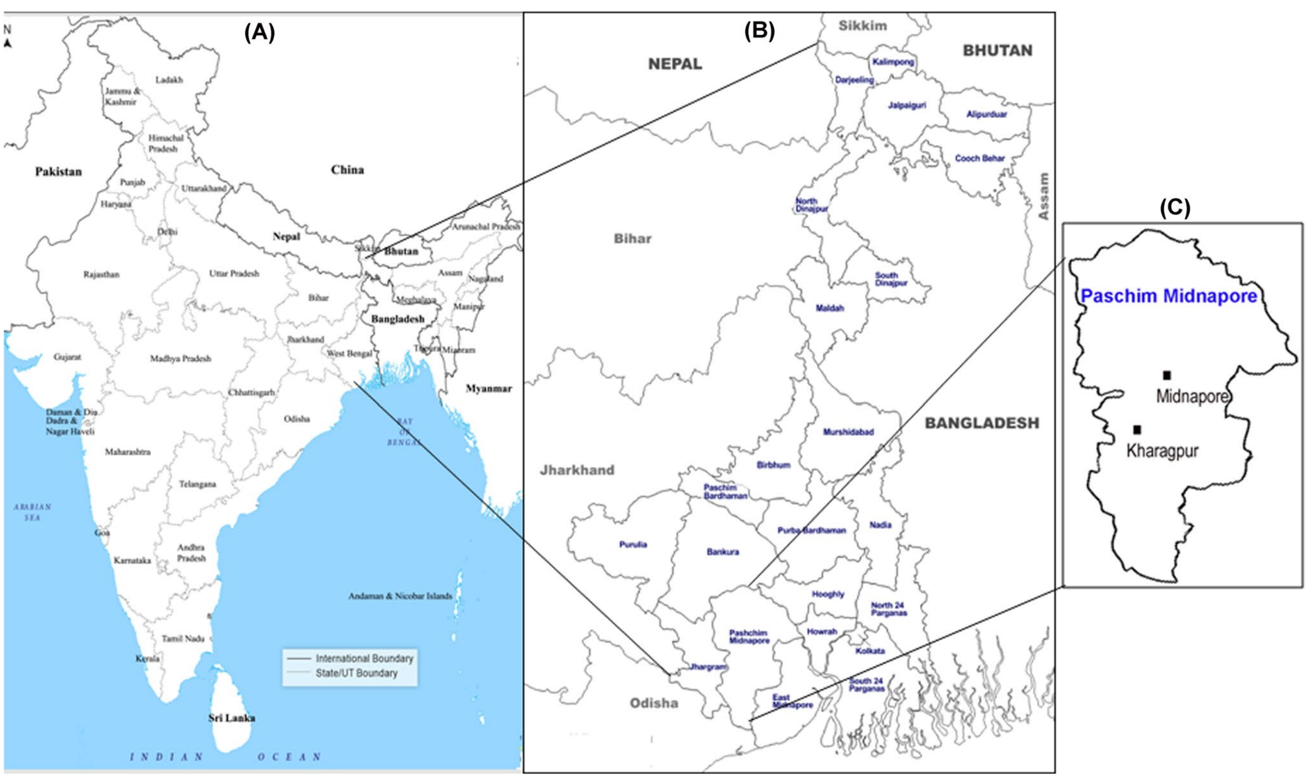


Fig. 1 The geographical location of Midnapore, the suburban area in eastern India. **a** The map of India, **b** the map of West Bengal, and **c** the location of Midnapore (study site) on the outskirt of Kharagpur

150 km from Kolkata, the state capital and India's seventh most populous city, and on the outskirts of Kharagpur, an important corridor to pan-India. SMH is the largest diagnostic service provider in Midnapore and a state-of-the-art 100-bed hospital under National Accreditation Board for Hospitals and Healthcare Providers (NABH) accreditation and ISO9001:2008 certification. Non-duplicate UPEC isolates were included in this study and recovered from symptomatic, uncomplicated male and female UTI cases of all age groups with significant bacteriuria attending outpatient departments (OPD) [1, 7, 15]. The isolates were excluded from patients with a history of vaginal symptoms, invasive urinary tract procedures, and/or hospitalization during the two weeks prior to the appearance of UTI symptoms. Patients were segregated into three age groups: pediatric (< 15 years), adults (15–64 years), and elders (≥ 65 years). Bacterial isolates were identified to species level by the VITEK2 automated system (bioMerieux,SA, Marcy l'Etoile, France) and stabbed into semi-solid nutrient agar (HiMedia, India) at SMH and transported to BMLVU for molecular characterization. This study was approved by the Institutional Review Board (IRB) of Vidyasagar University (Ref. VU/IEC/122/2020). Informed consent was waived because this study, with an observational nature, focused mainly on bacteria and involved no interventions with patients. The information of patients, including age, gender, and specimen, was shared in this study.

Molecular Identification

The stab cultures were revived onto eosin methylene blue agar (HiMedia) selecting green metallic-sheen colonies for molecular characterization. The isolates were preserved in brain heart infusion (BHI) broth (HiMedia) with 15% glycerol (v/v) at-70 °C for future investigation. DNA was extracted using the boiling lysis method for PCRs by emulsifying a single bacterial colony in 200 µl of molecular biology-grade water, incubating at 95 °C in a water bath for 10 min, chilling on ice for 5 min, and collecting the supernatant after centrifugation at 10,000 rpm for 10 min. Templates were stored at – 20 °C until used. Species-level identification was confirmed by PCR targeting *mdh* gene [14].

Resistotyping

The antibiotic susceptibilities of the isolates were determined by the Kirby Bauer disk diffusion method on Muller-Hinton agar (HiMedia) using seventeen antibiotics of eleven classes according to the Clinical and Laboratory Standards Institute (CLSI) guidelines: penicillin (ampicillin 10 µg), cephamycin (cefoxitin 30 µg), 3GC (ceftazidime 30 µg, cefotaxime 30 µg), carbapenem (meropenem 10 µg, imipenem 10 µg), fluoroquinolone (ciprofloxacin 5 µg, levofloxacin 5 µg), aminoglycoside (gentamicin 10 µg, amikacin 30 µg), sulphonamide (co-trimoxazole 1.25/23.75 µg), nitrofuran (nitrofurantoin 300 µg), phosphonic acid (fosfomycin 200 µg), and β-lactam/β-lactamase inhibitor (amoxicillin/clavulanic acid 20/10 µg; piperacillin–tazobactam 30/6 µg) [16]. Colistin susceptibility was assessed by the disk elution method on cation-adjusted Muller-Hinton broth at concentrations of 1, 2, and 4 µg/ml following the CLSI guidelines [16]. The agar dilution test was performed to determine the minimum inhibitory concentrations (MICs) of ceftazidime, cefotaxime, and meropenem at 2-fold serial dilutions from 2 to 256 µg/ml [16]. *Escherichia coli* ATCC25922, *Pseudomonas aeruginosa* ATCC27853, and *Staphylococcus aureus* ATCC25923 served as quality control strains. Results were interpreted as susceptible, intermediate, and resistant by the CLSI breakpoints [16]. Resistant and intermediate isolates were designated as non-susceptible. The MDR was defined as isolates showing non-susceptibility to at least one agent in ≥ 3 antimicrobial classes, while extensively drug-resistant (XDR) isolates are susceptible to ≤ 2 classes [17]. The isolates showing a zone of inhibition (ZOI) of ≤ 17 mm with either cefoxitin were suspected to be phenotypic AmpC producers, while ESBL production was screened by the combined disk diffusion with ceftazidime and cefotaxime alone and in combination with clavulanic acid, showing a difference of ≥ 5 mm [16].

Screening and Sequencing of β-lactamase and PMQR Genes

Sixteen major β-lactamase gene families were screened among the isolates by either end-point simplex or multiplex PCRs using published gene-specific primers [17, 18]. The β-lactamase families were as follows: penicillinase (OXA-1), AmpC cephalosporinases (ACC, DHA; MIR: ACT, CHE, and GC1; CMY-1: MOX and FOX; and CMY-2: CFE, LAT, and BIL), ESBLs (TEM, SHV, CTX-M-1, CTX-M-2, and CTX-M-9), and carbapenemases (KPC, NDM, VIM, IMP, and OXA-48). PMQR traits, including *qnrA, qnrB, qnrS, qepA*, and *aac(6')-lb-cr,* were detected by pentaplex-PCR as described earlier [19]. When using single DNA-extracted templates of each isolate, the results of the PCRs were validated by amplification of 16S rRNA in the carbapenemase panel. PCR amplification was carried out on Bio-Rad T100 (Bio-Rad, Hercules, California, US) and Master-cycler nexus gradient (Eppendorf, Hamburg, Germany) thermocyclers. Amplified products were analyzed, electrophoresed on a 2% pre-stained SeaKem LE agarose gel (Lonza Biosciences, US), and visualized by the Gel-Doc system (Bio-Rad). TEM, CTX-M, and NDM amplicons were purified using the NucleoSpin PCR Clean-Up Kit (DSS Takara Bio Inc., India) and outsourced to Barcode Biosciences Pvt. Ltd. (Karnataka, India) for Sanger sequencing with published primers [17, 18]. Gene variants were searched and matched by the NCBI-BLAST (https://blast.ncbi.nlm.nih.gov/Blast.cgi) and the EMBL-EBI program (https://www.ebi.ac.uk/about). The nucleotide sequences of TEM, CTX-M, and NDM alleles were submitted to the NCBI for GenBank accession numbers.

Plasmid Analysis

Plasmids were extracted from the overnight-grown Luria–Bertani (LB) broth (HiMedia) cultures by an alkaline-lysis method at elevated temperature using *Escherichia coli* V517 and *Shigella flexneri* YSH6000 as the reference strains [20]. The extracted plasmid was discriminated on 0.8% SeaKem LE agarose (Lonza, Basel, Switzerland) in 1X TAE buffer at 80 V for 2 h, followed by ethidium-bromide staining and visualization with the Gel-Doc system (Bio-Rad). Plasmid incompatibility (Inc) types FIA, FIB, FIC, FII, HI1, HI2, N, R, X1-X4, and ColE1 were identified by PCR-based replicon typing (PBRT) as described earlier [17].

Phylogrouping

The isolates were classified into one of eight phylogenetic groups (A, B1, B2, C, D, E, F, and G) or the cryptic clades I to V using a quadruplex-PCR assay [21, 22]. The presence of the *chuA* gene indicates the strains belonging to groups B2 and D, and its absence shows groups A and B1. The *yiaA* gene discriminates between groups B2 and D, while the *ybgD.1* gene differentiates between groups F and G [21, 22].

Genetic Fingerprinting

Bacterial isolates were subcultured overnight on LB broths following DNA extraction with the Wizard Genomic DNA Purification Kit (Promega, Madison, US). DNA quality and quantity were assessed by measuring absorbance with a BioPhotometer Plus instrument (Eppendorf) and stored at − 20 °C for further use. The isolates were fingerprinted using enterobacterial repetitive intergenic consensus (ERIC)-PCR with the ERIC2 primer as described earlier [17]. ERIC-PCR was performed in the Bio-Rad T100 thermocycler (Bio-Rad) by an initial denaturation at 94 °C for 5 min, followed by 35 cycles of denaturation at 94 °C for 45 s, annealing at 52 °C for 1 min, and extension at 68 °C 5 min with a final extension at 68 °C for 15 min. PCR products were separated with 1.5% (w/v) SeaKem LE agarose gel electrophoresis containing 0.5X-TBE buffer (SRL, India) at 80 V for 120 min containing ProxiO 1 kb DNA ladder (SRL) followed by ethidium-bromide staining and visualization by the Gel-Doc system (Bio-Rad).

Strain Typing

The phylogroup B2 isolates were screened for sequence type (ST) 131 by simplex-PCR as described earlier [23]. *Escherichia coli* multilocus sequence typing (MLST) was performed by amplifying internal fragments of seven housekeeping genes (*adk, fumC, gyrB, icd, mdh, purA*, and *recA*) and Sanger sequencing for determining the sequence types (STs) and clonal complexes (CCs) in the *Escherichia coli* MLST database website (http://mlst.warwick.ac.uk/mlst/dbs/Ecoli/documents/primersColi_html) [24].

Data Analysis

Graphical representation and statistical analysis were performed using Microsoft Office Excel 2010 and GraphPad Prism software version 7.0 for Windows. The $\chi 2$ test was used to compare the categorical variables at a *p*-value cutoff of <0.05, indicating a statistically significant value. The MAR index was estimated as the ratio of the number of antibiotics to which an isolate was resistant to the total number of antibiotics tested [25]. A MAR index >0.2 indicates a high-risk of contamination with antibiotic-resistant bacteria originating from a source of large amounts of antibiotic use. Plasmid size was estimated with *Escherichia coli* V517 (plasmid sizes: 54 kb, 5.6 kb, 5.1 kb, 3.0 kb, 2.7 kb, and 2.1 kb) and *Shigella flexneri* YSH6000 (212 kb) as the reference strains using Image Lab software ver. 5.2.1 for Windows (Bio-Rad). Extracted plasmids were designated as small (≤25 kb), medium (>25–50 kb), large (>50–100 kb), and mega (>100 kb) size [17]. A dendrogram was constructed from ERIC-PCR fingerprint bands normalized against a 1 kb DNA ladder using GelJ software ver. 2.0 for Windows [17]. Similarity analysis and clustering of matched bands were calculated using the Dice coefficient with 1.5% tolerance and the unweighted pair group method with arithmetic mean (UPGMA), respectively. Each cluster of the dendrogram was alphabetically assigned. Clonal relatedness was spotted in isolates showing ≥80% similarity, while the ERIC sub-types were marked with ≥90%.

Results

Demographic Information and Bacterial Isolates

A total of 102 non-duplicate *mdh*-positive UPEC isolates were recovered during the study period (2021, $n=16$; 2022, $n=51$; and 2023, $n=35$) from CAUTI patients with a median age of 31 years (range, 5 months to 91 years). The patients belonged to the following age groups: pediatrics ($n=16$, 15.7%), adults ($n=49$, 48%), and elders ($n=37$, 36.3%). The ratio of males to females was 1:1.5.

Resistotypes

Of the first-line antibiotics, fosfomycin and nitrofurantoin were susceptible in all (100%) and 76 (74.5%) isolates ($n=102$), respectively (Fig. 2). Carbapenems ($n=88$, 86.3%) had the highest in vitro activity following aminoglycosides (gentamicin: $n=66$, 64.7% and amikacin: $n=72$, 70.6%) and co-trimoxazole ($n=54$, 52.9%) as alternative antibiotics. Ampicillin ($n=9$), fluoroquinolones ($n=18$), and 3GCs (ceftazidime: $n=20$ and cefotaxime: $n=30$) were found least effective ranging 10–30% (Fig. 2). Among the β-lactam/β-lactamase inhibitors, piperacillin–tazobactam ($n=71$, 69.6%) was more effective than amoxicillin–clavulanic acid ($n=32$, 31.4%) for the isolates. Phenotypic AmpC and ESBL production were detected in 75 (73.5%) and 51 (50%) isolates, respectively. Interestingly, 37 (70%) ESBL-producing isolates remained susceptible to piperacillin–tazobactam. Two colistin-resistant isolates had MIC 4 µg/ml and the rest categorized as intermediate with MIC ≤ 1 µg/ml. Thirteen carbapenem-resistant *Escherichia coli* (CREC) co-resisted nitrofurantoin ($n=5$), amikacin ($n=10$), gentamicin ($n=9$), and/or co-trimoxazole ($n=12$). Eleven profiles were formed based on the susceptibility of 3GCs, carbapenems, and fluoroquinolones in the isolates, of which the majority (> 50%) belonged to 3GC-fluoroquinolone resistance (Table 1). Of the ceftazidime ($n=82$) and cefotaxime ($n=72$) non-susceptible isolates, the MICs reached ≥ 256 µg/ml for both 3GCs, with a median of 32 µg/ml and 64 µg/ml, respectively. Meropenem non-susceptible isolates ($n=13$) had maximum MICs 16 µg/ml (median 8 µg/ml). The MAR index ranged from 0 to 0.9 with a median of 0.5 among the isolates ($n=102$) of which 79 (> 75%) had > 0.2 (Table 2). Of the 78 MDR isolates, 73 (93.6%) had a MAR index > 0.2 (χ^2, 64.7; P, 0.0001) harboring up to seven tested ARGs (Table 2). Five XDR (< 5%) carried 4–6 ARGs with a MAR index of ≥ 0.8. Notably, five isolates were pan-susceptible having no ARGs (Tables 1 and 2).

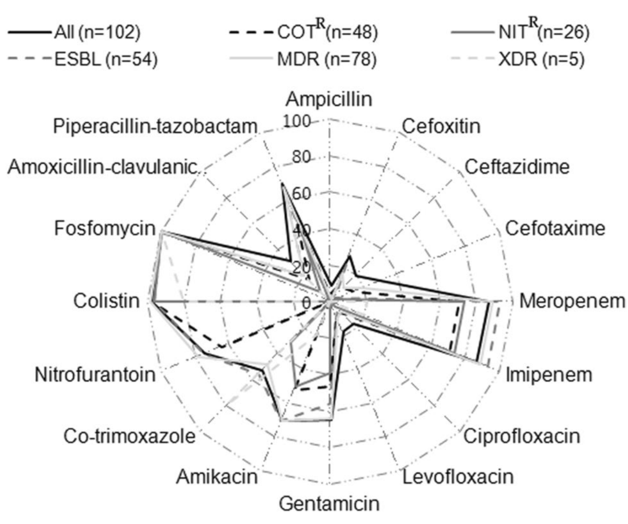


Fig. 2 Radar plot of antimicrobial susceptibilities among *Escherichia coli* isolates ($n=102$) causing community-acquired urinary tract infections in Midnapore, West Bengal

β-lactamase and PMQR Genes

The AmpC, ESBL, and PMQR genes were tested in all study isolates ($n=102$), while carbapenemases were screened in CREC ($n=13$). ESBL-producing *Escherichia coli* (ESBL-EC) isolates were predominated by CTX-M-1 ($n=52$, 51%) and TEM ($n=49$, 48%), followed by penicillinase (OXA-1: $n=24$, 23.5%), AmpC (CMY-2: $n=17$, 16.7%), and carbapenemase (NDM: $n=13$, 12.7%) (Table 1). Fourteen out of the sixteen representative isolates had TEM-1 variant (NCBI accession no.: PP465049, PQ143077-PQ143087), while two novel alleles were assigned as TEM-253 (ECUTI112, PQ037599) and TEM-254 (ECUTI116, PQ037600). CTX-M-15 and NDM-5 were noted in the representative eighteen (PQ143088-PQ143105) and six isolates (PQ156577-PQ156582), respectively. Other tested β-lactamases were rarely detected, such as DHA (7%), SHV (7%), and OXA-48 (2%) (Table 1). None had KPC, VIM, and IMP. Of the 88 β-lactam-resistant isolates, 83 (94.3%) carried up to five of the tested β-lactamases as follows: single β-lactamase ($n=32$, 38.6%), double ($n=31$, 37.3%), triple ($n=15$, 18.1%), tetra ($n=2$, 2.4%), and penta ($n=3$, 3.6%). Among the 79 ESBL-genotype confirmed isolates, twenty-three (29.1%) isolates co-carried TEM and CTX-M-1 (χ^2, 18.3; P, 0.0001), while three isolates (3.8%) contained TEM-CTX-M-1-SHV. Two isolates (ECUTI017 and ECUTI111) were double carbapenemase producers (NDM-5/OXA-48) co-harboring CMY-2/DHA/TEM and CMY-2/TEM/CTX-M-1, respectively. Strikingly, eight 3GC-susceptible isolates carried TEM, and one had either SHV or CTX-M-1. Five β-lactam-resistant isolates, even two 3GC-resistant, didn't harbor any of the tested β-lactamases. Of the 70 AmpC-, penicillinase-, and/or ESBL-carrying carbapenem-susceptible isolates, piperacillin–tazobactam ($n=50$, 71.4%) was more effective than ampicillin/clavulanic acid ($n=17$, 24.3%) (Table 3). Notably, piperacillin–tazobactam remained susceptible in 34 (76%) ESBL-producing isolates, but declined to 10 (56%) in the presence of OXA-1.

Of the PMQR genes, *aac(6')-lb-cr* ($n=24$, 23.5%) was prevalent, followed by *qnrS* ($n=14$, 13.7%) (Table 1). The *qnrA* and *qnrB* were carried by each isolate, while two had *qepA*. Notably, 42 fluoroquinolone non-susceptible isolates (Intermediate: 6 and resistant: 36) were devoid of any PMQRs. Three fluoroquinolone-resistant isolates carried double PMQR, either *aac(6')-lb-cr* with *qnrA* or *qnrS* and *qepA* with *qnrS*.

Table 1 Distribution of β-lactamases, PMQR genes, and plasmids in the isolates ($n = 102$) under different susceptibility profiles of third-generation cephalosporins, carbapenems, and fluoroquinolones

Susceptibility			Isolate (*n*)	β-lactamase (*n*)[b]							PMQR (*n*)[c]		Plasmid Inc type (*n*)[d]					Plasmid number; size range, kb[e]
3GC	CAR	FQ		CMY-2	DHA	OXA-1	TEM	SHV	CTX-M-1	NDM	*qnrS*	*aac*	FIA	FIB	FII	ColE1	R	
S	S	S	10[a]	1	NP	NP	3	1	1	NP	NP	NP	1	2	2	2	NP	0–4; 3.7–124.1 [7]
R	R	R	10	7	1	4	9	1	2	10	4	3	8	6	9	4	NP	2–8; 1.1–>212 [9]
R	R	IS	1	NP	NP	NP	1	NP	NP	1	NP	NP	NP	NP	1	NP	NP	2; 56.1–141.1 [1]
IS	S	S	1	NP	NP	NP	NP	NP	NP	NP	NP	NP	NP	1	1	NP	NP	1; 38.4 [1]
R	S	S	3	NP	NP	1	NP	NP	2	NP	NP	NP	NP	2	2	3	NP	4; 2.3–131.7 [2]
R	IS	R	2	1	NP	NP	2	NP	2	2	2	1	2	1	2	1	1	4; 2.1–212 [1]
R	S	IS	7	NP	1	NP	3	1	5	NP	2	NP	5	4	5	5	1	0–6; 2.1–212 [6]
IS	S	R	6	1	1	2	1	NP	NP	NP	NP	2	4	1	4	1	NP	4–6; 2.1–212 [5]
R	S	R	53	7	4	17	25	4	40	NP	5	18	29	20	38	12	4	0–9; <2.1–>212 [39]
S	S	IS	4	NP	NP	NP	2	NP	NP	NP	NP	NP	NP	2	2	1	NP	2; 2.2–212 [3]
S	S	R	5	NP	NP	NP	3	NP	NP	NP	1	NP	NP	2	4	NP	NP	0–6; 2.1–212 [4]
Total (*n*, %)			102 (100)	17 (16.7)	7 (6.9)	24 (23.5)	49 (48)	7 (6.9)	52 (50.9)	13 (12.7)	14 (13.7)	24 (23.5)	49 (48.0)	41 (40.2)	70 (68.6)	29 (28.4)	6 (6)	0–9; 1.1–>212 [78]

3GC third-generation cephalosporins, *CAR* carbapenems, *FQ* fluoroquinolones, *PMQR* plasmid-mediated quinolone resistance, *aac aac(6')-lb-cr*, *ND* not done, *NP* not present, *S* susceptible, *IS* intermediate-susceptible, *R* resistant

[a]Five out of ten isolates were pan-susceptible

[b]Three cefoxitin-resistant isolates belonging to $3GC^S$-CAR^S-FQ^S profile harbored CMY-2, TEM, SHV, and/or CTX-M-1. OXA-48 was carried by each of the two isolates of $3GC^R$-$CAR^{R/IS}$-FQ^R

[c]Each of four isolates of $3GC^R$-CAR^S-FQ^R profile contained *qnrA* ($n = 1$), *qnrB* ($n = 1$), and/or *qepA* ($n = 2$)

[d]IncFIC and IncX1 were present in each of one isolate of $3GC^R$-CAR^S-FQ^R profile

[e]Plasmid was extracted among representative isolates of each susceptibility profile as mentioned in parenthesis

Table 2 Phylogroup distribution across diverse multiple-antibiotic resistance indices of UPEC study isolates

MAR index ($n=102$)	Antibiotic resistance class/gene (range)	Phylogroup ($n=68$)[d]						
		A	B1	B2	C	D	F	Untype
0.0 (5, 4.9%)	0C/0G	3	–	1	–	–	–	–
0.1 (10, 9.8%)	1-2C/0-2G	2	–	–	–	–	–	1
0.2 (8, 7.8%)[a]	2-4C/0-2G	1	–	3	–	–	–	2
0.3 (6, 5.9%)	3-5C/0-1G	1	–	1	–	–	–	1
0.4 (10, 3.9%)	3-6C/0-2G	–	1	3	–	–	–	1
0.5 (24, 23.5%)	5-7C/1-4G	1	2	7	1	1	–	3
0.6 (19, 18.6%)	6-8C/1-4G	1	1	3	3	1	1	3
0.7 (6, 5.9%)	7-8C/1-4G	–	–	4	–	–	1	–
0.8 (13, 12.7%)[b]	8-9C/1-7G	1	–	3	1	2	1	2
0.9 (1, 1%)[b]	9C/4G	–	–	1	–	–	–	–
> 0.2 (79, 77.5)[c]	3-9C/0-7G	4	4	22	5	4	3	10

MAR multiple-antibiotic resistance index, *MDR* multidrug resistance

[a]5 out of 8 and all were MDR

[b]4 out of 13 isolates with MAR index 0.8 and one with 0.9 was/were XDR, respectively

[c]Of the 79 isolates, 73 and 5 were MDR and XDR, respectively

[d]Phylogenetic typing was performed among representative isolates of each MAR index group

Table 3 Activity of β-lactam/β-lactamase inhibitors among AmpC, penicillinase, and/or ESBL-producing carbapenem-susceptible UPEC isolates ($n=70$)

β-lactamase profile (*n*)	BL/BLI susceptible	
	AMC	PTZ
CMY-2 (2)	–	2
DHA (5)	–	4
OXA-1 (1)	–	1
TEM (11)	6	11
SHV (1)	–	1
CTX-M-1 (11)	5	7
SBLP (31)	11	26
CMY-2/TEM (3)	–	3
CMY-2/CTX-M-1 (3)	–	1
OXA-1/SHV (1)	–	1
OXA-1/CTX-M-1 (11)	1	6
TEM/SHV (1)	1	1
TEM/CTX-M-1 (11)	4	8
DBLP (30)	6	20
CMY-2/TEM/CTX-M-1 (1)	–	–
OXA-1/CTX-M-1/TEM (5)	–	2
TEM/SHV/CTX-M-1 (3)	–	2
TBLP (9)	–	4
Total (70)	17	50

BL/BLI β-lactam/β-lactamase inhibitor, *SBLP* single β-lactamase producer, *DBLP* double β-lactamase producer, *TBLP* triple β-lactamase producer

Plasmid Profiles

A total of 78 representative isolates, including 63 MDR and 5 XDR, were selected for plasmid analysis (Table 1). The isolates carried up to nine plasmids (median, 4) ranging in size from 1 to > 212 kb (Fig. 3). Mega plasmids were harbored by 48 (62%) isolates among which 45 (94%) were MDR. Thirteen isolates co-carried two mega-plasmids. Large, medium, and/or small plasmids were carried by 23 (29.5%), 20 (25.6%), and 47 (60.3%) isolates, respectively. Five isolates had 2–6 small plasmids (< 15 kb) which were positive for either OXA-1, TEM, CTX-M-1, or *aac(6')-lb-cr*.

Of the 102 isolates, 83 (81.4%) remained typable in precedence of the IncFII (68.6%), FIA (48%), FIB (40.2%), and ColE1 (28.4%) (Table 1). Twenty-nine (34.9%) isolates co-presented IncFIA and FIB, while one had IncFIC. Moreover, a significant correlation (χ^2, 42.7; *P*, 0.0001) was seen between IncFI and FII in fifty-eight isolates. The association between plasmid size and type, as well as gene localization, had yet to be investigated. IncF plasmids were found in 31 isolates containing mega-plasmids, except for one with small plasmids (< 15 kb). IncF was the only plasmid type in 71 (85.5%) isolates, with 65 (91.5%) having a MAR index > 0.2 and up to six ARGs. Notably, 41 ESBL-UPEC were typed exclusively for the IncF plasmid. Thirteen NDM-positive CREC consistently harbored both the IncFI and FII plasmids. Six isolates showing MAR index of ≤ 0.5 were positive for ColE1 only among which three had TEM, SHV, CTX-M-1, and/or *qnrB* co-existing untypable large and mega plasmids. IncR was found in six MDR isolates with a MAR index of 0.4–0.8, together with IncF ($n=5$) and/or ColE1 ($n=3$), whereas IncX1 was found in one MDR isolate

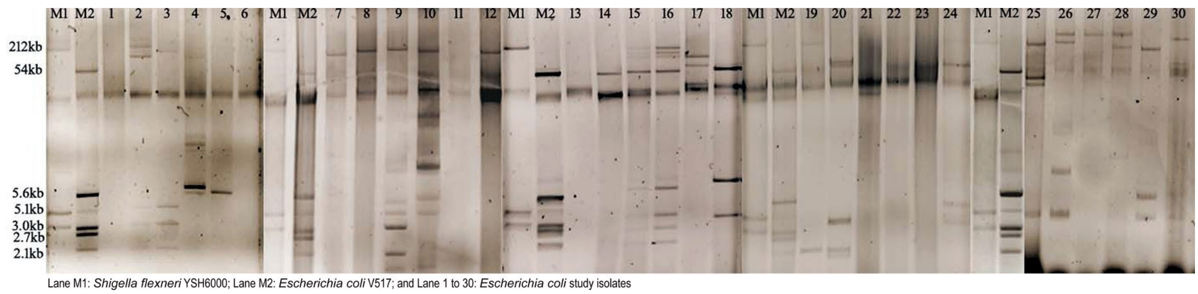


Fig. 3 Agarose gel electrophoresis of extracted plasmids from the representative study isolates

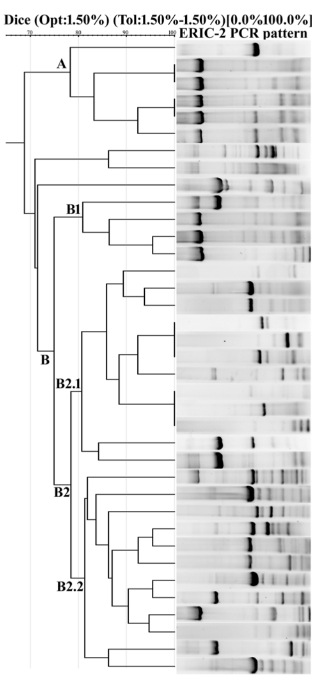
with a MAR index of 0.7. None had IncHI and N. Nineteen isolates were untyped, among which eleven carried 1–4 plasmids of 3–200 kb and nine had CMY-2 ($n=2$), DHA ($n=3$), OXA-1 ($n=3$), TEM ($n=1$), SHV ($n=1$), CTX-M-1 ($n=3$), and/or *aac(6')-lb-cr* ($n=3$). Among ten plasmid-free isolates, seven were MDR with a MAR index ≥ 0.5 harboring either CMY-2 ($n=2$), TEM ($n=4$), CTX-M-1 ($n=5$), or *aac(6')-lb-cr* ($n=2$) and IncFI ($n=4$), FII ($n=5$), R ($n=1$), or ColE1 ($n=2$) types. Five XDR isolates had 2–8 plasmids of IncFI/II type. One pan-susceptible isolate contained a 38 kb untypable plasmid without any β-lactamases or PMQRs (Table 1).

Phylogroups

A total of 68 representative isolates of different MAR index were phylogrouped (Table 2). Twenty-six isolates (> 38%) belonged to phylogroup B2 mostly MAR index > 0.2 ($n=22$) and carrying ESBLs ($n=20$). Ten had phylogroup A (15%) of which 6 showed MAR index (≤ 0.2). Phylogroups B1, C, D, and F were infrequently observed (< 8%), but all showed MAR index > 0.2. Interestingly, six out of thirteen CREC were belonged these groups. Of the eight CREC phylotyped isolates, six were non-B2 phylogroup (A: 1, D: 2, C: 2, and F: 1) and two remained untypable. Phylogroup D and F CREC isolates co-carried one PMQR trait with four to five β-lactamases (Fig. 4). Each of two CREC (MAR index 0.8) was phylotyped as D and F co-harboring one PMQR with either four or five β-lactamases.

Genetic Fingerprints and Strain Types

Thirty-seven UPEC representing various phylogroups generated 22 distinct ERIC types (E01 to E22) with 68%



Isolate ID	Patient	ERIC-type	Phylogroup	ST/CC	AMR profile	β-lactamase/PMQR pattern	Inc pattern
ECUTI027	E/F	E01	Untypable	ND	3GC/CAR/FQ/AMG/COT	NDM-5	Untypable
ECUTI110	E/M	E02	F	648/648	3GC/CAR/FQ/*AMG*/COT/NIT	CMY-2/OXA-1/TEM-1/CTX-M-15/NDM-5/aac(6')-Ib-cr	FIA/FIB/FII
ECUTI114	E/F	E02	B2	131/131	3GC/FQ/*AMG*/NIT	OXA-1/CTX-M-1/aac(6')-Ib-cr	FIA/FIB/FII
ECUTI118	E/F	E03	C	ND	3GC/FQ	CMY-2/CTX-M	Untypable
ECUTI133	A/F	E03	B2	131/131	3GC/FQ/AMG/COT/NIT	OXA-1/CTX-M-1/aac(6')-Ib-cr	Untypable
ECUTI127	A/F	E03a	B2	131/131	3GC/FQ/AMG	OXA-1/CTX-M-1/aac(6')-Ib-cr	Untypable
ECUTI123	A/F	E04	B2	131/131	PEN/*FQ*	TEM-1	FIB/FII
ECUTI126	A/F	E05	B2	131/131	3GC/*FQ*	CTX-M-15	FIA/FIB/FII
ECUTI105	P/F	E06	B2	131/131	3GC/FQ/AMG/COT/NIT	OXA-1/TEM/CTX-M-1/aac(6')-Ib-cr	FIA/FIB/FII
ECUTI099	P/M	E07	B2	non-131	3GC/*FQ*	CTX-M-15/qnrS	FIA/FIB/FII/ColE1
ECUTI119	A/M	E08	B2	131/131	*3GC*/FQ/NIT	OXA-1/aac(6')-Ib-cr	FIA/FII
ECUTI116	P/F	E09	B2	131/131	3GC/FQ	OXA-1/TEM-254/CTX-M-15/aac(6')-Ib-cr	FIA/FII
ECUTI037	A/M	E09a	B2	131/131	3GC/AMG/COT	OXA-1/CTX-M-1	FIB/FII/ColE1
ECUTI101	P/M	E10	Untypable	ND	3GC/*FQ*/AMG/COT	TEM-1/CTX-M-15	FIA/FIB/FII/ColE1
ECUTI031	A/F	E11	C	2851/23	3GC/*CAR*/FQ	TEM-1/CTX-M-15/qnrS	FIA/FII/ColE1
ECUTI152	A/F	E11a	Untypable	ND	3GC/FQ/*COT*	TEM/qepA	FIA
ECUTI022	E/M	E12	A	361/361	3GC/CAR/FQ/AMG/COT	CMY-2/TEM-1/NDM-5	FIA/FIB/FII
ECUTI103	E/M	E12	Untypable	ND	2GC	CMY-2	ColE1
ECUTI026	E/M	E12	Untypable	ND	3GC/CAR/FQ/AMG/COT	CMY-2/OXA-1/TEM-1/NDM-5/qnrS	FIA/FIB/FII
ECUTI143	A/F	E12a	B2	non-131	3GC	CTX-M-15	FIB/FII/ColE1
ECUTI153	P/F	E13	Untypable	ND	3GC/FQ/COT	CMY-2/TEM-1/CTX-M-15	FIA/FII
ECUTI096	E/F	E13	Untypable	ND	*3GC*/FQ	CMY-2	Untypable
ECUTI091	P/M	E13	B2	131/131	3GC/FQ/AMG/COT/NIT	OXA-1/TEM-1/CTX-M-15	FII/ColE1
ECUTI034	E/M	E14	B2	non-131	*3GC*/AMG/COT	NP	FIB/FII
ECUTI085	P/M	E15	B1	ND	3GC/*FQ*	CTX-M-15	FII
ECUTI131	P/M	E16	B2	non-131	*3GC*/FQ/COT/NIT	DHA	FIA/FII
ECUTI092	A/F	E17	B2	non-131	FQ	NP	FIA/FIB/FII/ColE1
ECUTI145	P/F	E18	Untypable	ND	3GC/FQ/AMG/*NIT*	TEM/CTX-M-1	FIA/FII
ECUTI140	A/M	E19	B1	ND	3GC/FQ/COT	TEM/qnrS/qepA	FIA
ECUTI112	P/F	E19a	A	ND	PEN	TEM-253	FIB/FII
ECUTI040	E/F	E19a	B1	ND	3GC/FQ/COT	CMY-2/CTX-M-15/aac(6')-Ib-cr	R
ECUTI129	A/M	E20	C	ND	3GC/FQ	CMY-2/CTX-M-1	FIA/FII
ECUTI136	A/F	E20a	F	405/405	3GC/FQ/COT/NIT/COL	CTX-M-15	FIA/FII/ColE1
ECUTI121	A/F	E20a	B2	131/131	3GC/FQ/*AMG*/NIT/*COL*	OXA-1/CTX-M-15/qnrA/aac(6')-Ib-cr	FIA/FIB/FII
ECUTI014	A/M	E20a	D	405/405	3GC/CAR/FQ/AMG/COT	OXA-1/TEM-1/CTX-M-15/NDM-5/aac(6')-Ib-cr	FIA/FIB/FII
ECUTI083	P/M	E21	B1	ND	3GC/FQ	CTX-M-1	FIB/FII
ECUTI076	A/F	E22	B2	131/131	3GC/FQ/AMG/COT/*NIT*	OXA-1/CTX-M-15/aac(6')-Ib-cr	FIA/FIB

Fig. 4 Dendrogram for the cluster analysis of thirty-seven uropathogenic *Escherichia coli* isolates ($n=37$) by ERIC-PCR during 2021–23

similarity, segregating clusters A ($n=6$, 16.2%) and B ($n=28$, 75.7%) (Fig. 4). Cluster A included six isolates with 78% identity, four of which were plasmid-free (E1, E3, and E3a). Cluster B was divided into two subclusters, B1 ($n=4$) and B2 ($n=24$), with B2 forming two sub-sub-clusters, B2.1 ($n=12$) and B2.2 ($n=12$). The isolates from clusters B1, B2.1, and B2.2 were clonally linked (> 80% similarity). Clusters B2.1 and B2.2 yielded five and seven different ERIC types. There was no clear link found between ERIC types and phylogroups, AMR, or plasmid profiles. Limited genetic diversity was noted among the prevalent ESBL-UPEC. Cluster A and B1 were mostly populated by ST131/phylogroup B2 isolates, while cluster B2 was non-ST131/phylogroup B2. Two *qepA*-carrying isolates were found different ERIC types (E11a and E19) of which one belonged to B1, while one *qnrA*-harboring isolate (E20a) of ST131 was found clonally related with ST405 isolates (Fig. 4). NDM-5-carrying ERIC types were clustered under A (E01, E02), B2.1 (E12), and B2.2 (E20a).

Of the 26 phylogroup B2 isolates, twenty-one isolates (80.8%) were belonged to ST131 commonly harbored CTX-M-1 and/or OXA-1. Nineteen ST131 isolates (90.5%) were MDR, while one was XDR. IncFI and/or IncFII plasmids were carried by eighteen ST131 isolates (85.7%). Five strain types (STs) were determined mainly targeting non-phylogroup B2 ESBL-EC and/or CREC as follows: ERIC-type/phylogroup/ST/clonal-complex/β-lactamase/PMQR/Inc: E20a/F/405/405/CTX-M-15/FIA.FII.ColE1; E20a/D/405/405/OXA-1.TEM-1.CTX-M-15.NDM-5/*aac(6')-Ib-cr*/FIA.FIB.FII; E12/A/361/361/CMY-2.TEM-1/NDM-5/FIA.FIB.FII; E02/F/648/648/CMY-2.OXA-1.TEM-1.CTX-M-15.NDM-5/*aac(6')-Ib-cr*/FIA.FIB.FII; and E11/C/2851/23/TEM-1.CTX-M-15.NDM-5/*qnrS*/FIA.FII.ColE1 (Fig. 4).

Discussion

In the era of AMR, UPEC accounts for around 40% of global deaths significantly impeding the progress of several Sustainable Development Goals (SDGs), importantly good health and well-being (SDG3) [8]. ESBL-EC is considered under the SDG3.d.2 to achieve the target by 2030. Recently, *Escherichia coli* flagged as the dominating uropathogen at both community (> 60%) and hospital (> 35%) settings in India [4]. Here, we addressed the crisis of AMR in a suburban population of eastern India through CAUTI by *Escherichia coli* during 2021–2023.

Patient demographic information indicated that female adults (> 35%) were mostly affected due to CAUTI in this study. Earlier observations marked 50–60% occurrence of UTI among adult women in a lifetime [1, 2, 7, 26]. AMR shifts the empirical treatment of common infections toward "Watch" and even "Reserve" group antibiotics disobeying the antimicrobial stewardship, particularly in community [1, 5, 8]. Fosfomycin (100%) and nitrofurantoin (75%) well served as first-line antibiotics for treating acute cystitis in this region, while susceptibility to alternative antibiotics (carbapenems: 90–86%, aminoglycosides: 86–65%, and co-trimoxazole: 93–53%) were marginally to moderately declined since 2006 (Fig. 5) [26–28]. Strikingly, we observed up to > 25-fold rise of resistance to earlier drugs-of-choice 3GCs (3–80%) and fluoroquinolones (7–82%) (Fig. 5) [26–28]. Piperacillin–tazobactam remained active in > 70% of the infections, especially as anti-ESBL and

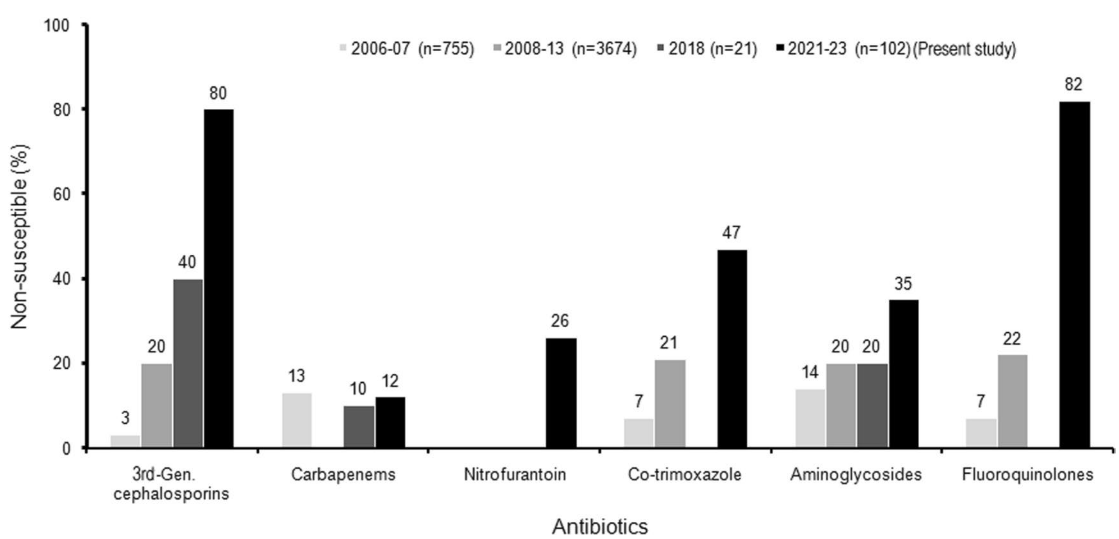


Fig. 5 Antibiotic resistance trend among uropathogenic *Escherichia coli* since the last two decades in Midnapore, West Bengal

carbapenem-sparing agent [29]. Colistin resistance (< 2%) was first noted in this region. In addition of high resistance (> 80%) to 3GCs, fluoroquinolones, and co-trimoxazole, carbapenem resistance was recorded among 60% of UPEC isolated from hospitalized patients in Kolkata, India, during 2010–2011 [30]. In the DASH multi-centric mentorship-based pan-India study during 2022, Rizvi et al. [7] represented almost similar antibiograms of *Escherichia coli* causing CAUTI across 22 sites as follows: fosfomycin (92–97%), nitrofurantoin (69–97%), meropenem (81–94%), piperacillin–tazobactam (65–93%), gentamicin (67–90%), co-trimoxazole (37–59%), 3GCs (29–85%), and ciprofloxacin (23–48%) [7]. In 2023, the ICMR reported lesser susceptibility in nosocomial UPEC isolates than community isolates based on surveillance network of 21 hospitals across India especially for intravenous antibiotics: ceftriaxone (70% and 85%), meropenem (15% and 15%), amikacin (10% and 25%), and piperacillin–tazobactam (35% and 52%) [4]. Interestingly, the marginal difference was noted in empirical antibiotics like fosfomycin (< 4%), nitrofurantoin (10–15%), co-trimoxazole (50–60%), and for reserve-antibiotic colistin (< 1%) [4]. Overall, clinicians relied on fosfomycin and nitrofurantoin rather than co-trimoxazole and aminoglycosides for treating CAUTI due to increasing resistance to 3GCs and fluoroquinolones in LMICs, including India [1, 5, 7]. In contrast, fosfomycin resistance reached up to 17% among hospitalized UTI Czechs-patients in Central Europe during 2013–2018 [31]. Several developed nations control the AMR following stringent antimicrobial stewardship, even 3GCs (< 40) and fluoroquinolones (< 30%) [5].

Majority of the study isolates (> 80%) were MDR with MAR index of > 0.2 indicating the antibiotic misuse and high-risk pathogen exposure in this region. We detected a significant proportion of food-producing animal-derived MDR *Escherichia coli* resistant to 3GCs and fluoroquinolones, even colistin in this geographic region (unpublished data). This could be owing to the mis/over-use of antibiotics for treatment, prophylaxis, and growth promotion of livestock and poultry, resulting in high and cross-resistance and raising the risk superbug infection among community individuals in LMICs [8]. Other Indian and international studies have hinted the risk of cross-infection with MDR isolates with MAR index of up to 0.8 at community and hospital levels [3, 30, 32].

ESBL-UPEC carrying CTX-M-1 and TEM (> 50%) was common in this study co-harboring OXA-1 (> 20%), CMY-2 (> 15%), and/or NDM (> 10%). This study recorded a rare occurrence (< 10%) of SHV, DHA, and OXA-48. South Asia is the epicenter of ESBL-EC, which occurs up to 80%, including India [7, 13, 30, 33]. TEM and CTX-M co-occurred in 29% of the ESBL-producing study isolates which were rarely (4%) associated with SHV. Since last two decades, the prevalence of SHV was gradually declined with the emergence of CTX-M in *Escherichia coli* [33–35]. TEM-1 and CTX-M-15 alleles were found in the study isolates, which supported previous community and hospital reports [33, 36, 37]. Low-to-moderate dispersion (< 30%) of OXA-1, CMY-2, SHV, DHA, and OXA-48 were occurred through UPEC in India and Bangladesh [37, 38]. The elevated prevalence of ESBL-EC confronts difficult-to-treat and difficult-to-control UTIs, even in nations with excellent public health and healthcare [32, 38]. In this study, *aac(6')-lb-cr* was mostly detected (> 20%) followed by *qnrS* (> 10%) with rare findings of *qnrA, qnrB,* and *qepA*. Interestingly, Kammili et al. [33] observed higher *qnrS* (> 60%) than *aac(6')-Ib-cr* (> 35%) without *qnrA/B/D* in ESBL-EC CAUTIs among primigravid women in India [33]. PMQR and ESBLs were present in > 45% of hospital UPEC Indian isolates with a high co-occurrence (> 35%) of OXA-1, TEM, CTX-M, and *aac(6')-Ib-cr* [30]. Prevalence of *aac(6')-Ib-cr* up to 65% followed by *qnrS* (> 40%) and *qnrB* (> 20%) was documented in *Escherichia coli* from China, Canada, Egypt, and Germany, while *qnrA and qepA* were reported in China, France, and Japan [6, 12]. Co-production of β-lactamases and PMQRs up to six genes was noted in > 20% study isolates. However, > 10% β-lactam-resistant and > 45% fluoroquinolone-resistant isolates lacked the ARGs evaluated. Apart from plasmidic genes, chromosome-mediated mechanisms including efflux activity and porin loss also impart AMR in *Escherichia coli* [5, 12, 17, 19].

Extra-chromosomal genetic entity triggers intra- and inter-species spread of numerous ARGs via horizontal gene transmission in bacterial populations, resulting in drug-resistant infections [39]. The majority (> 90%) of the studied isolates contained plasmids, importantly mega and large (> 75%). Large plasmids, such as IncF, are commonly involved in carrying and catering the ARGs among bacterial pathogens resulting rapid dissemination of AMR [17, 37, 39]. IncF plasmid was present in > 70% of the study isolates, especially with MDR phenotypes. IncFI and/or IncFII plasmids were also prevailed among MDR-UPEC isolates carrying ESBL, PMQR, and even carbapenemase genes from other community as well as hospital study sites in India [30, 34, 36, 38]. The correlation between ARGs and plasmid size and type were yet to be established in this study. We noticed the ColE1 plasmid in 25% of isolates harboring ARGs, including CTX-M-1. ColE1 belongs to small plasmid that contributes in bacterial evolution and pathogenesis rather than AMR [17]. Jan et al. [40] noted the prevalence of MDR-UPEC isolates (> 50%) in India carried single or multiple small plasmids ranging from 2 to 26 kb [40]. Seven plasmid-free MDR study isolates contained ARGs and Inc-types. Chromosomal integrations of ARGs and plasmids occasionally occur, conserving the AMR for a protracted duration in bacterial cells as a reservoir and deciding to excise for dissemination upon stimulation with antibiotics [41].

Bacterial forensic research, such as phylogrouping and strain typing, is critical in public health because it reveals population structure, ecological niches, pathogenicity, and AMR in a particular geographic region, allowing for strategic containment. Phylogroup B2 (38%) was the most frequent in the studied community, with 78% producing ESBL (Table 2). The remaining six phylogroups (A, B1, C, D, E, and F) were found in 5–10% of the study isolates. The B2 and D are sister groups confounding in virulent extra-intestinal isolates, while the A and B1 are sister groups prevalent as intestinal pathogens and commensal isolates [42]. Other studies marked the prevalence of B2 (> 40%) among ESBL-UPEC followed by D (> 20%) and B1 (> 10%) irrespective of community or hospital settings [42–44]. Surprisingly, 70% of the study isolates belonging to A, B2, C, and D caused UTIs in females. These phylogroups are classified as cervico-vaginal *Escherichia coli* (CVEC), which poses a risk of chorioamnionitis in pregnant women, and group D, which has the potential to cause urosepsis [45]. Chakraborty et al. [43] reported D as the second most common type among UPEC in south India [43]. Two studies revealed the predominance of A in Indian and Egyptian UPEC carrying ESBL and carbapenemase [38, 44]. Phylogroup F study isolates (n = 3) were MDR with MAR index ≥ 0.6, consistent with prior findings [22].

The study isolates produced 22 ERIC types with a diversity of 68% (Fig. 4). The isolates were clonally related under each of four (sub) clusters showing eight clones (E02, E03, E09, E11, E12, E13, E19, and E20). UPEC ERIC fingerprints were akin to earlier rare findings in India and Nigeria [10, 11, 30, 35]. In this study, ST131 was detected in > 80% of phylogroup B2 isolates associated with MDR phenotype mostly carrying IncF plasmid and CTX-M-15 (Fig. 4). ST131 coupled with B2 is the third most epidemic clone of ExPEC isolates causing UTI as well as bacteramia in both community and hospital settings among humans and food-producing animals [9, 10, 34]. MDR ST131 clone, primarily resistant to fluoroquinolones and 3GCs, disseminates CTX-M-15 horizontally by IncF plasmid or vertically prevailing in America, Europe, and Oceania over the last ten years [46]. Noticeably, the clone is occasionally observed in South and Southeast Asia, including India [34]. ERIC types demonstrated clonal relatedness (similarity ≥ 80%) among study ST131 isolates (Fig. 4). Hussain et al. also revealed the clonal evolution of Indian UPEC ST131 isolates by pulsotyping in 2012 [34]. Two clonally related (E20a) isolates of phylogroup F and D were ST405 carrying CTX-M-15 and/or NDM-5. ST405 linked with group D is the fifth most common global ESBL and carbapenemase-producing clone widely present in India [9]. ST361, ST648, and ST2851 were assigned to phylogroups A (E12), F (E02), and C (E11), respectively. These STs had up to six ARGs, notably NDM-5. ST361 and ST648 are global emerging clones of carbapenem–colistin-resistant *Escherichia coli* with NDM-5/mcr-1 that infect both humans and animals [9, 47]. ST2851 was seldom reported from India, Bangladesh, and Switzerland, having the ability to transmit ESBLs and carbapenemases [9].

Conclusions

To the best of our knowledge, this is the first molecular epidemiological investigation on AMR through the critical priority pathogen *Escherichia coli* causing CAUTI at a suburban area located on the outskirts of Kharagpur, an important corridor to pan-India. Several epidemic clones of *Escherichia coli*, resistant to 3GCs, carbapenems, and even colistin, infiltrate in this community. Strikingly, the prevalence of phylogroup B2 MDR-UPEC ST131 clone is a potential public health threat dispersing ESBLs, especially CTX-M-1. The study's multimodal aspects could be utilized by clinicians for empiric treatment, microbiologists for resistance traits, and epidemiologists for surveillance applications to comprehend and contain the regional spread of AMR. The molecular outcomes could be harnessed for downstream strategic genomic surveillance to clarify the source trace and map of MDR epidemic clones in this region, limiting their cross-border transmission at both the national and international levels.

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Author Contributions UB: Format analysis, writing draft, editing, validation, and evaluation; AM: Format analysis and evaluation; TS: Review and editing; SubD: Format analysis and resources of urine culture isolates; SurD: Conceptualization, supervision, methodology, writing original draft, and editing. All authors have read and approved the final version.

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Data Availability The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest No potential conflict of interest was reported by the author(s).

Ethical Approval and Consent to Participate This study was approved by the Institutional Review Board (IRB) of Vidyasagar University (Ref. VU/IEC/122/2020). Informed consent was waived because this study, with an observational nature, focused mainly on bacteria and involved no interventions with patients.

Consent for Publication Not applicable.

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